Hence, $5.0 - F = 3.0 \Rightarrow F = 2.0$ N

6 D

Let T be the time of the ascent and v be the speed of the ball in the last second of the ascent. Let h be the height of the ascent

Using $v^2 = u^2 + 2as$,

$$0^2 = v^2 - 2g(0.25h)$$

$$\Rightarrow \quad v^2 = 2g(0.25h) = 0.50gh$$

Using $v = u + at$,

$$0 = \sqrt{0.50gh} - g \times 1$$

$$\Rightarrow h = 19.62 \text{ m}$$

Let the initial speed at the bottom of the throw be u.

Using $v^2 = u^2 + 2as$, $0^2 = u^2 - 2 \times 9.81 \times 19.62$

$$\Rightarrow u = 19.62 \text{ m s}^{-1}$$

Using $v = u + at$, $0 = 19.62 - 9.81 \times T$

$$\Rightarrow T = 2.00 \text{ s}$$

7 A